

Conditioning and Directionality in Adversarial Transfer on CIFAR-10

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Abstract

How sensitive are conclusions about adversarial transfer to direction, perturbation budget, and source-success conditioning? We evaluate FGSM and projected-gradient attacks on the full CIFAR-10 test set using three independently trained ResNet-18, VGG16, and MobileNetV2 checkpoints. The corrected evaluation excludes existing clean errors and reports both pairwise transfer (PTR) and transfer conditional on source success (CTR), with exact denominators. At 8/255, mean PGD transfer ranges from 18.09% for ResNet-to-VGG to 95.23% for VGG-to-MobileNet, revealing strong and seed-stable directionality. Reducing the budget to 2/255 lowers those rates to 5.84% and 23.29%, respectively. Conditioning has its largest effect when the source attack is imperfect: for VGG-sourced PGD at 2/255, CTR exceeds PTR by 5.97–8.68 percentage points across the two targets. A seed-0 100-step sensitivity check changes every off-diagonal 2/255 PTR by at most 0.27 points relative to the 40-step, five-restart protocol. These results show that transfer claims can change materially with direction, budget, and evaluation population even when labels, preprocessing, and threat models are held fixed.

1 Introduction

Adversarial examples expose a gap between ordinary predictive accuracy and behavior under deliberately chosen perturbations. FGSM provides a gradient-based construction [1], while iterative optimization enables broader searches within a threat model [2, 3]. Transfer attacks use a source model to construct an input that is subsequently evaluated on another model. The target supplies neither gradients nor feedback during construction.

Transferability is an empirical property of a source, a target, an attack, and an evaluation population. It cannot be inferred from a high error rate alone. If the target already misclassifies most clean inputs, an unchanged input may appear to be a successful attack under an unconditional metric. A mismatch between label spaces makes this problem more severe.

The study asks whether conclusions about attack rankings and model-pair differences are stable under different,

explicitly stated conditioning populations. The intended empirical contribution is a reproducible characterization of this measurement sensitivity across training seeds and attack settings. The self-correction in Section 3 motivates this question; it is not presented as evidence of a general failure rate across the literature. The implementation and metric definitions support the study, but neither their existence nor a large observed rate difference alone establishes a novel empirical finding.

2 Background and scope

Goodfellow et al. connect adversarial vulnerability to high-dimensional linear behavior and introduce FGSM [1]. Madry et al. formulate robustness using optimization over a perturbation set [2]. Carlini and Wagner develop optimization-based attacks and demonstrate the importance of careful evaluation [3]. Liu et al. investigate transfer and black-box attacks [4]. These studies motivate examining source attack strength separately from target transfer. None establishes that one attack must universally transfer better than another.

Carlini et al. systematize the methodological foundations and common pitfalls of adversarial-robustness evaluation [7]. Croce and Hein introduce AutoAttack, an ensemble intended to reduce dependence on attack tuning and improve evaluation reliability [8]. Tramer et al. emphasize defense-specific adaptive evaluation and illustrate how apparently adequate evaluations can miss weaknesses [9]. RobustBench standardizes robustness benchmarking under specified tasks and threat models and uses AutoAttack as part of its evaluation framework [10].

Our study complements these evaluation efforts by examining source-to-target transfer and the populations on which transfer rates are conditioned. It does not introduce a stronger attack, certify robustness, or replace adaptive defense evaluation. The FGSM/PGD study does not constitute an AutoAttack evaluation; citing AutoAttack and RobustBench must not be interpreted as having run either benchmark. Attack-strength sensitivity and the source white-box controls are necessary for interpreting the transfer measurements.

The present protocol concerns untargeted digital attacks against deterministic image classifiers. It does not

evaluate physical attacks, detection systems, adversarially trained defenses, or deployment security. CIFAR-10 results would not by themselves justify claims about autonomous vehicles or face recognition. The selected architectures are established CNN baselines, not a comprehensive representation of contemporary vision systems.

3 Motivating case study: a self-correction

The audit refers to repository commit 58f8102 and its four experiment scripts. The accompanying manuscript states that models were trained on CIFAR-10 and reached approximately 90% clean accuracy. The released scripts instead instantiate ImageNet-pretrained torchvision models, without CIFAR-10 training or replacement of their 1,000-class output heads. They compare the resulting output indices directly with CIFAR-10 labels numbered 0 through 9. The two sets of indices do not share the required class semantics.

3.1 Preprocessing and budgets

The transfer scripts feed raw CIFAR-10 tensors to the ImageNet models without their expected preprocessing. The figure script applies a different normalization and measures its budget in that transformed space. Thus, its perturbation size is not directly interchangeable with a budget in raw pixels. The PGD script uses a relative step size of 0.01, rather than the manuscript’s stated absolute step of $2/255$. The CW call uses only ten optimization steps, which does not establish convergence. Its L_2 budget must also be distinguished from the L_∞ budgets of FGSM and PGD.

3.2 Sampling and success definitions

The table scripts count every target error on attacked inputs as transfer success, without checking whether either model was initially correct. They do not preserve a record of source success. Some experiments select an unseeded subset of 1,000 images, despite broader full-test-set descriptions. Repeated source-target calls regenerate attacks rather than explicitly reusing one recorded source perturbation across all targets.

These observations establish a mismatch between the published explanation and the released artifact. They do not reconstruct unavailable training histories or prove the origin of every previously reported table entry. No compatible checkpoints or per-example result logs were present in the audited repository. The earlier rates and architecture rankings are therefore excluded from the revised empirical record.

4 Why error is not attack success

Let f_t be the target predictor, y the true label, and $A_s(x)$ an attack constructed using source s . Define clean error $e_t = \Pr[f_t(x) \neq y]$, newly induced error $q = \Pr[f_t(A_s(x)) \neq y \mid f_t(x) = y]$, and persistence of existing error $r = \Pr[f_t(A_s(x)) \neq y \mid f_t(x) \neq y]$. By partitioning on clean correctness,

$$\Pr[f_t(A_s(x)) \neq y] = (1 - e_t)q + e_tr. \quad (1)$$

For the identity attack, $q = 0$ and $r = 1$, so adversarial error equals clean error. An illustrative classifier with 99% clean error therefore has 99% adversarial error under an identity attack and zero newly induced errors. This is an algebraic example, not a measurement from CIFAR-10.

The unconditional attacked-error rate and CTR generally refer to different populations. Their difference is therefore not an isolated estimate of inflation caused by clean errors, and it need not be positive. The term e_tr directly measures the fraction of all inputs that were already wrong and remain wrong after attack; $(1 - e_t)q$ measures newly induced errors on the same all-input denominator. The signed difference between unconditional attacked error and q equals $e_t(r - q)$. These identities explain the error attribution that motivates clean-error exclusion; PTR and CTR additionally impose source-correctness and source-success conditions. Undefined conditional rates remain undefined when their conditioning population is empty.

4.1 Recorded estimands

For a fixed evaluation set of N examples, let $C_s(i)$ and $C_t(i)$ indicate clean correctness. Let $S_s(i)$ and $S_t(i)$ indicate misclassification of the source-generated attacked input. Define the pairwise eligible set $E_{st} = \{i : C_s(i) \wedge C_t(i)\}$ and successful-source subset $F_{st} = \{i \in E_{st} : S_s(i)\}$. We report

$$\text{ASR}_s = \frac{\sum_{i:C_s(i)} S_s(i)}{\sum_i C_s(i)}, \quad (2)$$

$$\text{PTR}_{st} = \frac{\sum_{i \in E_{st}} S_t(i)}{|E_{st}|}, \quad (3)$$

$$\text{CTR}_{st} = \frac{\sum_{i \in F_{st}} S_t(i)}{|F_{st}|}. \quad (4)$$

PTR measures target failure on jointly clean-correct inputs whether or not the source attack succeeded. CTR measures transfer conditional on source success. Both are needed: a high CTR can describe a small, selected subset. The diagonal is a white-box control; its CTR is tautologically one whenever its denominator is positive. It must not be interpreted as black-box evidence.

All fractions include numerator and denominator counts. An empty denominator is undefined, not zero.

Clean accuracy and attacked accuracy on the full evaluation set are also reported. Wilson 95% intervals describe uncertainty under a binomial sampling interpretation for fixed checkpoints; they are descriptive when evaluating the entire finite CIFAR-10 test set and do not measure variability across training runs. Pairwise eligibility sets differ, so raw rates across model pairs are not strictly matched-population comparisons.

5 Corrected experimental protocol

5.1 Data and model training

The pipeline uses the standard CIFAR-10 class order [5]. A fixed split seed of 1729 partitions the 50,000 training images into 45,000 training and 5,000 validation images. Validation data receive no random augmentation. Training uses random crops with four-pixel padding and random horizontal flips. The 10,000 test images are reserved for final evaluation; checkpoint selection uses validation accuracy only.

All networks have ten output classes and are initialized without ImageNet weights. The ResNet-18 stem uses a 3×3 convolution at stride one and no initial max pool. VGG16 uses its convolutional feature stack, global 1×1 pooling, and a single 512-to-10 linear classifier. MobileNetV2 changes its initial convolution to stride one. These adaptations, especially the compact VGG classifier, must be stated when comparing with other implementations.

Inputs to the attack interface are floating-point pixels in $[0, 1]$. Channel normalization with mean (0.4914, 0.4822, 0.4465) and standard deviation (0.2470, 0.2435, 0.2616) occurs inside the model, so gradients include preprocessing while perturbation budgets remain in raw-pixel coordinates. Checkpoints contain architecture, class order, split indices, training seed, selected epoch, and model state. Evaluation rejects incompatible checkpoint metadata.

Training uses 200 epochs, SGD with momentum 0.9, weight decay 5×10^{-4} , cosine learning-rate decay, and batch size 128. ResNet-18 and MobileNetV2 use initial learning rate 0.1. That rate caused the VGG16 loss to remain near $\log 10$ and validation accuracy near chance, so we stopped that failed run and selected 0.01 using validation performance only. We then froze 0.01 for all three VGG16 seeds before test evaluation. The independently trained seeds are 0, 1, and 2; the split seed remains 1729.

5.2 Attacks and controls

FGSM uses one signed cross-entropy gradient step with projection into the pixel range. PGD uses random initialization within the intersection of the pixel range and the L_∞ ball, 40 steps, absolute step size $2/255$, and five

restarts at the primary budget $8/255$. Across iterates and restarts, the implementation prioritizes source misclassification, then larger source loss. Target predictions never influence this selection. This source-selected construction may differ from final-iterate PGD in its transfer behavior, so the selection rule is part of the attack specification.

The clean control leaves inputs unchanged. The random-noise control samples independent uniform perturbations within the same L_∞ budget and clips to the pixel range. The empirical scope is limited to the L_∞ threat model. We do not mix CW- L_2 results with these measurements or imply that equal numeric radii across norms represent equal threat models.

For every source and attack, the same generated inputs are evaluated on every target, including the source. Models remain in evaluation mode. Runtime checks reject nonfinite pixels, invalid pixel ranges, and norm violations. A saved example records its test index, true label, predictions, and actual perturbation. Visualization does not presume attack success or establish perceptual invisibility.

5.3 Reproducibility and sensitivity analysis

A seeded permutation specifies test indices, and the identical ordered sample list is used for every pair. The final study should use the full test set. A 1,000-example subset is suitable for a clearly labeled pilot only. Each completed run saves arguments, software versions, commit identifier, checkpoint SHA-256 hashes, test indices, elapsed time, clean and attacked per-example predictions, and per-example L_2 and L_∞ norms. Tables are generated from measured summaries rather than embedded constants.

We evaluate L_∞ radii $2/255$, $4/255$, and $8/255$. The PGD step size is one quarter of the radius. At $8/255$, seed 0 uses 10, 40, and 100 steps with one restart and 100 steps with five restarts; seeds 1 and 2 repeat the 10-step, one-restart and 100-step, five-restart endpoints. At $2/255$, seed 0 also uses 100 steps and five restarts. Attack hyperparameters are not selected to maximize target transfer. We show training seeds individually and summarize seed-level estimates; we do not pool repeated test images as independent observations.

6 Results

6.1 Clean accuracy and primary attack strength

Table 1 reports validation-selected checkpoint performance on the untouched test set. Variation across the three training seeds is small relative to the differences among architectures. At the primary $8/255$ budget, 40-step, five-restart PGD attains 100% source ASR for every ResNet-18 and MobileNetV2 checkpoint and at least

Table 1: Clean test accuracy (%) by training seed. The final column is the mean and sample standard deviation over three checkpoints.

Model	Seed 0	Seed 1	Seed 2	Mean $\pm$ SD
ResNet-18	95.11	95.01	95.20	95.11 $\pm$ 0.10
VGG16	90.59	90.41	90.88	90.63 $\pm$ 0.24
MobileNetV2	89.97	90.57	90.38	90.31 $\pm$ 0.31

Table 2: PGD PTR (%) at 8/255, reported as the mean [minimum, maximum] over three training seeds. Rows are sources and columns are targets. Diagonal entries are white-box controls and are omitted.

Source	ResNet-18	VGG16	MobileNetV2
ResNet-18	—	18.09 [16.73,19.38]	72.27 [71.76,72.73]
VGG16	92.73 [92.40,92.99]	—	95.23 [94.80,95.61]
MobileNetV2	67.60 [63.46,70.61]	25.94 [25.73,26.29]	—

99.98% for every VGG16 checkpoint. The diagonal therefore confirms that the primary attack is strong against its source models.

Uniform random noise at the same 8/255 radius produces mean off-diagonal PTR between 1.45% and 7.89%, below the corresponding adversarial results. Noise CTR is based on much smaller source-success subsets and is not used to rank model pairs. Because each nominal source row uses a separately seeded noise realization, these controls establish a non-adversarial baseline within each row rather than a paired comparison among noise rows.

6.2 Directionality at the primary budget

Table 2 shows off-diagonal PGD PTR at 8/255. Transfer is strongly directional. VGG16 examples transfer to ResNet-18 at 92.73% on average, whereas the reverse direction reaches 18.09%. MobileNetV2-to-VGG16 is 25.94%, whereas VGG16-to-MobileNetV2 is 95.23%. The ordering persists across all three seeds, so it is not produced by one checkpoint draw. These are associations among the trained architectures and recipes; differences in capacity, optimization, and clean accuracy preclude an architecture-only causal interpretation.

6.3 Budget and conditioning sensitivity

Transfer falls sharply with the perturbation budget, but the directional ordering is broadly stable (Table 3). VGG16-to-ResNet-18 drops from 92.73% at 8/255 to 15.45% at 2/255; ResNet-18-to-MobileNetV2 drops from 72.27% to 21.61%. These changes rule out treating one reported transfer percentage as a property of a model pair independent of its threat model.

At 2/255, VGG16 source ASR is only 71.05–72.98%. Conditioning on successful source attacks raises mean transfer from 15.45% PTR to 21.42% CTR toward ResNet-18, and from 23.29% to 31.97% toward Mo-

Table 3: Mean PGD PTR (%) over three seeds as a function of L_∞ budget.

Direction	2/255	4/255	8/255
ResNet-18 $\rightarrow$ VGG16	5.84	9.65	18.09
ResNet-18 $\rightarrow$ MobileNetV2	21.61	42.40	72.27
VGG16 $\rightarrow$ ResNet-18	15.45	54.52	92.73
VGG16 $\rightarrow$ MobileNetV2	23.29	64.50	95.23
MobileNetV2 $\rightarrow$ ResNet-18	10.79	31.80	67.60
MobileNetV2 $\rightarrow$ VGG16	4.61	10.69	25.94

bileNetV2. The corresponding gaps are only 0.37 points or less for ResNet-18 and MobileNetV2 sources, whose source ASRs are at least 94.15% and 99.82%. At 8/255, near-perfect source ASR makes PTR and CTR nearly identical. Conditioning therefore changes the interpretation most when source optimization fails on a substantial part of the jointly clean-correct population.

The stronger 100-step, five-restart 8/255 endpoint changes off-diagonal PTR by at most 1.70 percentage points relative to the primary 40-step, five-restart setting across all seeds. Increasing optimization does not increase every transfer entry: for example, ResNet-18-to-VGG16 falls slightly. At 2/255 on seed 0, increasing from 40 to 100 steps at five restarts changes all off-diagonal PTR entries by at most 0.27 points and CTR entries by at most 0.40 points. This supports convergence of the primary step count without asserting a global optimum.

FGSM reinforces the distinction between source strength and transfer. At 8/255, its mean PTR is 27.80% for ResNet-18-to-VGG16 and 29.34% for MobileNetV2-to-VGG16, compared with 18.09% and 25.94% for PGD. PGD nevertheless has far higher source ASR. For the other four off-diagonal directions, PGD transfers more strongly than FGSM. An attack ranking therefore depends on the source-target direction and cannot be inferred from white-box source success alone.

7 Evidence status and limitations

The final record contains nine checkpoints, full-test predictions, exact eligibility counts, perturbation norms, manifests, and checkpoint and source hashes. The three seed triads provide evidence about variability under the specified training procedure, but three draws support descriptive ranges rather than precise population-level inference. Wilson intervals in the generated per-run tables describe fixed-checkpoint binomial variation and do not capture training variability. The same 10,000 test images recur across seeds, so image-level counts must not be pooled as 30,000 independent observations.

The study covers three standard CNN families on CIFAR-10 and untargeted L_∞ attacks. It does not establish behavior on larger datasets, transformers, adversarially trained models, targeted attacks, or physical settings.

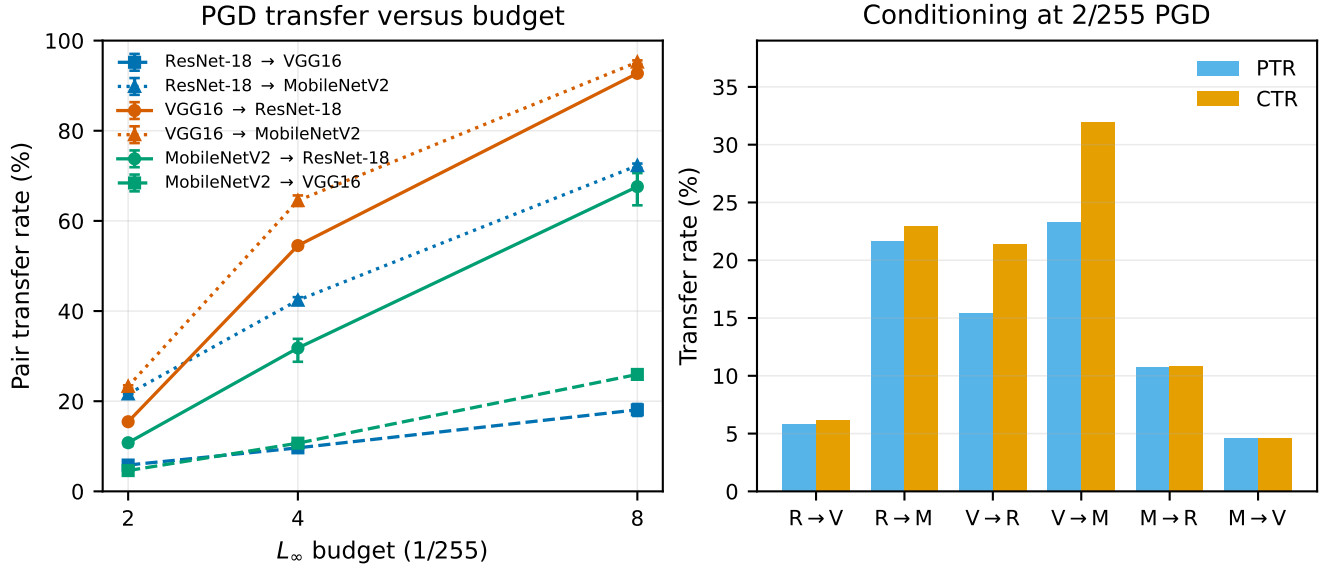


Figure 1: Left: mean PGD PTR over three seeds at each budget, with minimum-to-maximum seed bars. Line color denotes the source and marker/line style denotes the target. Right: mean PTR and CTR for PGD at 2/255; R, V, and M denote ResNet-18, VGG16, and MobileNetV2.

It also does not compare against every modern transfer attack. The VGG16 learning-rate adjustment was selected using validation results after the common recipe failed; this documented architecture-specific choice may itself affect comparisons. Random-noise draws are independently generated for each nominal source row, so noise-control rows are reproducible but do not share one common perturbation realization. Finally, the run manifests record a dirty working tree. Their source-file hashes identify the executed implementation, but a clean tagged release and archived artifacts would provide a stronger provenance record.

The original artifact is a motivating self-correction, not evidence that similar errors are common elsewhere. The present results do not imply that one architecture is intrinsically more transferable: architecture, parameterization, training dynamics, and clean accuracy vary together. PTR and CTR answer different questions, and neither should be selected after seeing which produces the more dramatic result.

8 Conclusion

Across three training seeds, adversarial transfer on CIFAR-10 is strongly directional and budget dependent. At 8/255, mean PGD PTR spans 18.09–95.23% across off-diagonal pairs; at 2/255, it spans 4.61–23.29%. Source-success conditioning changes rates most when the source attack is imperfect, producing PTR-to-CTR increases of up to 8.68 points for VGG16-sourced 2/255 PGD. Stronger source optimization can also reduce transfer for a particular direction, so source ASR and target transfer

must be reported separately. Compatible labels and pre-processing make these measurements meaningful; clean-error exclusions, explicit denominators, seed-level reporting, and convergence checks make their limits visible.

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